Register No. : | | | | | | | | | | | QP-Code : 20EC11T

I Semester Diploma Examination

DIGITAL ELECTRONICS (20EC11T)

Exam Date / Time: 20th Sep. 2023 / 2.00 PM

Time: 3 Hours **Max.Marks: 100**

Instructions: (1) Answer **one** full question from each section.

(2) One full question carries **20** marks.

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SECTION – I

1 (A). (i) Compare Analog signal and Digital signal 5

(ii) Convert decimal number $(50)_{10}$ into binary. 3

(iii) Represent hexa decimal number $(2EA4)_{16}$ in binary. 2

(B). (i) List the types of number system. Explain binary number system 5

(ii) Subtract $(1001)_2$ from $(1101)_2$. 2

(iii) Add $(21A)_{16}$ and $(1E4)_{16}$. 3

2 (A). (i) Explain Excess-3 code with an example 5

(ii) Subtract 1001 from 1100 using 2's complement method 5

(B). (i) Explain commutative law 5

(ii) State and prove De-morgan's theorem. 5

SECTION – II

3. a) Define Logic gate. Write symbol, Truth table and logic expression of AND, OR and NOT gates. (10)

b) Realise AND, OR , NOT, NAND and EX-OR gates using NOR gates. (10)

4. a) Simplify the following Boolean expression using K-map and draw the logic diagram. (5)

F(A,B,C)=A'B'C'+A'B'C+ A'BC'+ A'BC+ ABC'+ ABC

b) Convert the following (5)

i) SOP to POS, $F(A,B,C)= \Sigma(1,3,5,6)$

ii) POS to SOP, $F(A,B,C)= \pi(0,1,2,4)$

c) Simplify the following using Boolean laws and draw the logic diagram. (10)

i) y=(A+B)(A+B'')(A'+B)

ii)y=ABCD+AC(CD)'+(AB)'CD

SECTION – III

5. (a) (i) Explain half subtractor with truth table. 5

(ii) Write the comparison between serial and parallel adder. 5

(b) Explain working of a full adder with logic diagram and truth table. 10

6. (a) Explain working of 3 bit parallel adder circuit. 10

(b) Explain two bit magnitude comparator with truth table and gate level circuit. 10

SECTION – IV

7. a) Define Multiplexer. Explain 2:1 Multiplexer with logic expression and truth table. (5)

b) Realise 4:1 Multiplexer using 2:1 multiplexer and explain its working. (10)

c) List the applications of Multiplexers. (5)

8. a) Realise AND and OR gates using 2:1 Multiplexer. (5)

b) List the applications of Encoder. (5)

c) Explain the Decimal to BCD Encoder with Truth table, equation and logic diagram. (10)

SECTION – V

9. (a) Sketch and explain logic circuit of BCD to decimal decoder. 10

(b) (i) Explain the working of 4-line to 2 line Encoder. 5

(ii) Mention any 5 applications of Decoder. 5

10. (a) (i) Mention any 5 advantages of IC's. 5

(ii) Mention any 5 features of standard TTL. 5

(b) (i) Describe the interfacing between TTL and CMOS. 5

(ii) List any 5 applications of Integrated Circuits. 5

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